Square-Root Normal Forms for a Fixed-Shift Type-II Gate: Recentered Ramanujan Rows, a Near-$11/21$ Factorable Wedge, and a Cross-Spectral Angular Criterion

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Abstract

Fix $h \neq 0$, let $W \in C_c^\infty((0,\infty);\mathbb{R})$, and put

$$\Lambda_R(n) = \sum_{q \le R} \frac{\mu(q)}{\varphi(q)} c_q(n), \qquad r_R = \Lambda - \Lambda_R, \qquad R = X^{1/2-\delta}.$$

TPC-15 reduced the weighted fixed-shift prime-pair problem to a broad Vaughan Type-II packet by using only rows of modulus at most R. We resolve two further layers of that packet.

First, the complete-period mean of $j \mapsto \Lambda_R(mj+h)$ is computed for every m, including $m > R$. The prime progression density minus this model mean is the explicit cutoff drift

$$\Delta_{h,R}(m) = \sum_{\substack{q \mid m \\ q > R}} \frac{\mu(q) c_q(h)}{\varphi(q)}.$$

The aggregate drift is small, although it is generally nonzero row by row. Maximal Bombieri–Vinogradov after this recentering moves the resolved source-row wall from R to

$$M = X^{1/2} \exp(-\sqrt{\log X}).$$

An asymmetric Vaughan identity then gives the square-root normal form

$$\sum_n \Lambda(n)\Lambda(n+h)W(n/X) = \mathfrak{S}(h) X I_0(W) - \sum_{\substack{\ell > M \\ k > 1}} \Lambda(\ell) r_R(\ell k + h) W(\ell k/X) + O_A(X(\log X)^{-A}).$$

Thus the left unbalanced wing is closed up to the classical square-root distribution wall. The right wing is not removed: a fixed-k slice is itself a two-linear-form Hardy–Littlewood problem.

Second, opening the original coefficient $\beta_V(k) = \sum_{d \mid k, d \le V} \mu(d)$ reveals the true progression modulus $q = \ell d$. All terms with $\ell d \le R$ are peeled off. On the remaining three-scale region, Maynard's fixed-residue theorem for factorable moduli transfers to an actual hard subpacket. In particular, for $\delta > 1/42$ with a fixed interior margin, a smooth packet near

$$D = X^{1/21}, \qquad L = X^{10/21}, \qquad K = X^{11/21}$$

is $o(X)$. This deletes one nonempty D-resolved wedge, not a whole (L, K) block.

Finally, we prove

$$\sum_n r_R(n)^2 F(n/X) = X \log(X/R) I_0(F) + O_{F,\delta}(X),$$

so the calibrated residual remains large in energy. The hard packet is an exact translation matrix coefficient and a specified Fourier coefficient of a cross-spectrum. Explicit norm bounds turn the missing estimate into a quantitative angular-decorrelation corridor, while phase averages enjoy unconditional power cancellation. No fixed-shift prime-pair asymptotic, twin-prime lower bound, new level of distribution, or breach of sieve parity is claimed.

Keywords: fixed-shift prime correlations; Ramanujan sums; Vaughan identity; Bombieri–Vinogradov theorem; factorable moduli; Type-II sums; translation spectrum; sieve parity.

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1 Introduction and theorem ledger

The fixed-shift correlation

$$\mathcal{P}_h(X;W) = \sum_{n\geq 1} \Lambda(n)\Lambda(n+h)W(n/X) \tag{1}$$

has conjectural main term $\mathfrak{S}(h)XI_0(W)$, where

$$I_0(W) = \int_0^\infty W(t)\,dt \tag{2}$$

and

$$\mathfrak{S}(h) = \prod_p \frac{1-\nu_p(h)/p}{(1-1/p)^2}, \qquad \nu_p(h) = \#\{a \pmod p : a(a+h) \equiv 0 \pmod p\}. \tag{3}$$

For an odd h the factor at $p=2$ vanishes. For a nonzero even h the series is positive. In particular, an asymptotic of the expected form at $h=2$ and a nonnegative nonzero W would imply infinitely many twin primes. This paper does not prove such an asymptotic.

TPC-15 [11] introduced the finite Heath–Brown Ramanujan model

$$\Lambda_R(n) = \sum_{q\leq R} \frac{\mu(q)}{\varphi(q)} c_q(n), \qquad r_R(n) = \Lambda(n) - \Lambda_R(n), \tag{4}$$

and, for fixed $0<\delta<1/2$, took

$$R = \left\lfloor X^{1/2-\delta} \right\rfloor, \qquad U = V = \left\lfloor R^{1/2} \right\rfloor. \tag{5}$$

The resulting Vaughan coefficient is

$$\beta_V(k) = \sum_{\substack{d\mid k\\ d\leq V}} \mu(d), \tag{6}$$

and the remaining packet is

$$\mathcal{H}^{\mathrm{HB}}_{h,R;U,V}(X) = \sum_{\substack{\ell>U\\ k>V}} \Lambda(\ell)\beta_V(k) r_R(\ell k+h) W(\ell k/X). \tag{7}$$

TPC-15 proves

$$\mathcal{P}_h(X;W) = \mathfrak{S}(h)XI_0(W) - \mathcal{H}^{\mathrm{HB}}_{h,R;U,V}(X) + O_A\big(X(\log X)^{-A}\big). \tag{8}$$

The negative sign in (8) will be retained throughout.

The broad support

$$X^{1/4-\delta/2} < \ell, k \ll_W X^{3/4+\delta/2}$$

is not yet a canonical description of the gate. Two issues are hidden in that display. First, the finite model has an exact progression mean even when the row modulus exceeds R; only its equality with the prime density fails. Second, after (6) is opened, the modulus seen by prime-distribution theorems is ℓd, not ℓ or k. The paper resolves these two issues separately.

1.1 Results

The first result computes the missing row mean. Define

$$\rho_h(m) = \mathbf{1}_{(m,h)=1}\frac{m}{\varphi(m)}, \qquad \rho_{h,R}(m) = \sum_{\substack{q\mid m\\ q\le R}} \frac{\mu(q)c_q(h)}{\varphi(q)}. \tag{9}$$

The second quantity is the exact complete-period mean of $j \mapsto \Lambda_R(mj+h)$ for every m. Their difference is

$$\Delta_{h,R}(m) = \rho_h(m) - \rho_{h,R}(m) = \sum_{\substack{q\mid m\\ q>R}} \frac{\mu(q)c_q(h)}{\varphi(q)}. \tag{10}$$

Although it need not vanish for $m > R$, its weighted aggregate up to the Bombieri–Vinogradov wall is $o(1)$ after the natural $1/m$ normalization.

Let

$$M = M(X) = \left\lfloor X^{1/2}\exp(-\sqrt{\log X})\right\rfloor. \tag{11}$$

The recentered row theorem in section 3 controls every bounded row mask up to M. Taking the asymmetric Vaughan parameters $U = M, V = 1$ gives

$$\mathcal{P}_h(X;W) = \mathfrak{S}(h)XI_0(W) - \mathcal{H}^{\sharp}_{h,R;M}(X) + O_A(X(\log X)^{-A}), \tag{12}$$

where

$$\mathcal{H}^{\sharp}_{h,R;M}(X) = \sum_{\substack{\ell>M\\ k>1}} \Lambda(\ell)r_R(\ell k+h)W(\ell k/X). \tag{13}$$

This is the square-root normal form. It closes the left source-factor wing $\ell < X^{1/2-o(1)}$. It does not close the opposite wing in which k is small and ℓ is large.

Returning to the symmetric packet, opening β_V and grouping by the row modulus $m = \ell d$ peels off all terms with $\ell d \le R$. The unresolved coefficient is therefore supported in the exact hyperbolic wedge

$$\ell d > R. \tag{14}$$

This three-scale formulation allows a genuine local advance. A smoothed form of Maynard [6, Theorem 1.1] controls fixed-residue prime errors for moduli $q = q_1q_2$ when

$$Q_1Q_2^2 < X^{1-\eta}, \qquad Q_1^{12}Q_2^7 < X^{4-\eta}, \qquad Q_1^{20}Q_2^{19} < X^{10-\eta}. \tag{15}$$

With $q_1 = d \asymp D$ and $q_2 = \ell \asymp L$, this theorem, together with the explicit drift (10), proves that one smooth hard subpacket near

$$D = X^{1/21-o(1)}, \qquad L = X^{10/21-o(1)}, \qquad K = X^{11/21+o(1)} \tag{16}$$

is $o(X)$. This is an imported large-modulus estimate transferred to a new packet; it is not a new level-of-distribution theorem. Since d is resolved, the conclusion does not apply to the complete (L, K) block.

The last part is spectral but uses only standard translation dynamics. For nonnegative $F \in C_c^\infty((0,\infty))$,

$$\sum_n r_R(n)^2F(n/X) = X\log(X/R)I_0(F) + O_{F,\delta}(X). \tag{17}$$

Thus r_R is correlation-calibrated but not small in ℓ^2. The packet (13) is an exact matrix coefficient of the integer translation and hence a specified Fourier coefficient of a cross-spectrum. The remaining condition is directional cancellation, not decay of total residual energy. Phase averaging proves strong cancellation for generic factor phases, but not at the unmodulated phase.

1.2 Claim ledger

Proved here	Not proved here
Exact finite-model row mean for every modulus	A fixed-h prime-pair asymptotic
Recentered Type-I closure up to $X^{1/2-o(1)}$	Cancellation of the full right wing
Asymmetric square-root Vaughan normal form	Cancellation of every dyadic block
Peeling of the region $\ell d \le R$	A new Bombieri–Vinogradov or Maynard range
$o(X)$ on one Maynard-admissible D-resolved wedge	$o(X)$ on the complete balanced (L, K) block
Residual energy law and translation cross-spectrum identity	A spectral gap for primes or a Riemann-zero realization
Phase-averaged factor-mask bounds	Cancellation at the distinguished zero phase

The distinction between substantial Type-I/II input and actual prime detection is central to the parity problem; see D. H. J. Polymath [2], Ford and Maynard [3], Selberg [8]. Nothing below asserts that the parity barrier has been crossed. The purpose is to replace one broad unnamed remainder by a sharper sequence of positive regions, exact drifts, and explicitly quantified gates.

2 Finite Ramanujan rows beyond the cutoff

The Ramanujan sum is

$$c_q(n) = \sum_{\substack{a \pmod q \\ (a,q)=1}} \mathrm{e}(an/q). \tag{18}$$

The model in (4) is periodic with period $\mathrm{lcm}(1, \ldots, R)$. We first calculate its mean on every row, without assuming that the row modulus is visible below R.

Lemma 2.1 (Complete-period mean at arbitrary modulus)**.** *For every $m \ge 1$, the complete-period mean of $j \mapsto \Lambda_R(mj + h)$ is*

$$\rho_{h,R}(m) = \sum_{\substack{q \mid m \\ q \le R}} \frac{\mu(q) c_q(h)}{\varphi(q)}. \tag{19}$$

Moreover,

$$\rho_h(m) = \sum_{q \mid m} \frac{\mu(q) c_q(h)}{\varphi(q)}. \tag{20}$$

Consequently (10) holds, and $\Delta_{h,R}(m) = 0$ whenever $m \le R$.

Proof. For a fixed q, averaging the exponential definition gives

$$\mathbb{E}_j c_q(mj + h) = \begin{cases} c_q(h), & q \mid m, \\ 0, & q \nmid m. \end{cases} \tag{21}$$

Indeed, for a reduced numerator a, the phase amj/q has mean zero unless $q \mid m$. Inserting (21) into the finite model proves (19).

The right side of (20) is multiplicative in m. At a prime $p \mid m$, its local factor is

$$1 + \frac{\mu(p) c_p(h)}{\varphi(p)} = \begin{cases} p/(p-1), & p \nmid h, \\ 0, & p \mid h. \end{cases}$$

Their product is exactly $\mathbf{1}_{(m,h)=1} m/\varphi(m) = \rho_h(m)$. Subtracting the truncated sum gives the drift formula. $\square$

The drift is pointwise small only after one pays for the number of divisors, but its averaged form is considerably cleaner.

Lemma 2.2 (Pointwise and aggregate cutoff drift)**.** *For fixed $h \neq 0$ and $2 \le R \le M$,*

$$|\Delta_{h,R}(m)| \ll_h \frac{\tau(m)\log\log(3m)}{R}, \tag{22}$$

$$\sum_{m\le M} \frac{|\Delta_{h,R}(m)|}{m} \ll_h \frac{\log(2M)\log\log(3M)}{R}. \tag{23}$$

Proof. The standard bound $|c_q(h)| \le (q,h) \le |h|$ and $q/\varphi(q) \ll \log\log(3q)$ give

$$\frac{|c_q(h)|}{\varphi(q)} \ll_h \frac{\log\log(3q)}{q}.$$

Every term in the tail has $q > R$, which proves (22). For the aggregate estimate, write $m = qr$ and discard the condition $q \mid m$ only after making that substitution:

$$\begin{aligned}\sum_{m\le M} \frac{|\Delta_{h,R}(m)|}{m} &\le |h| \sum_{R<q\le M} \frac{1}{q\varphi(q)} \sum_{r\le M/q} \frac{1}{r} \\ &\ll_h \log(2M) \sum_{R<q\le M} \frac{\log\log(3q)}{q^2}.\end{aligned}$$

The last tail is $O(\log\log(3M)/R)$. $\square$

We next record the finite-row formula that turns the complete-period mean into a smoothed progression estimate. The finite model has the divisor expansion

$$\Lambda_R(n) = \sum_{\substack{d\mid n\\ d\le R}} \lambda'_R(d), \qquad \sum_{d\le R} |\lambda'_R(d)| \ll R(\log(2R))^2, \tag{24}$$

proved in TPC-15. The second bound is deliberately nonoptimal but is uniform and sufficient here. We shall also use its pointwise form

$$|\lambda'_R(d)| \ll \frac{d}{\varphi(d)} \log(2R/d) \ll (\log(2R))^2. \tag{25}$$

Lemma 2.3 (Finite-row Ramanujan leakage)**.** *Let $F \in C_c^1((0,\infty))$. Uniformly for all $m, R \ge 1$,*

$$\sum_{j\ge 1} \Lambda_R(mj+h)F(mj/X) = \rho_{h,R}(m)\frac{X}{m} I_0(F) + O_F\big(R(\log(2R))^2\big). \tag{26}$$

The same conclusion holds for a uniformly bounded family of compactly supported C^1 weights, with the implied constant depending only on a common support and C^1 bound.

Proof. Insert (24). For a fixed d, put $g = (d, m)$. The congruence $d \mid mj + h$ is insoluble unless $g \mid h$; when it is soluble, it is one residue class modulo d/g. Euler summation on this lattice gives

$$\sum_{\substack{j\ge 1\\ d\mid mj+h}} F(mj/X) = \mathbf{1}_{g\mid h} \frac{g}{d}\frac{X}{m} I_0(F) + O_F(1).$$

The continuous coefficients recombine into the complete-period mean in lemma 2.1. Summing the lattice errors with (24) proves the result. $\square$

Combining lemmas 2.1 and 2.3 gives the exact bookkeeping identity

$$\sum_j r_R(mj+h)F(mj/X) = \Delta_{h,R}(m)\frac{X}{m}I_0(F) + \mathcal{E}_{m,h}(X;F) + O_F\big(R(\log(2R))^2\big), \tag{27}$$

where

$$\mathcal{E}_{m,h}(X;F) = \sum_j \Lambda(mj+h)F(mj/X) - \rho_h(m)\frac{X}{m}I_0(F). \tag{28}$$

Thus every row has three logically distinct pieces: an explicit cutoff drift, a prime-progression error, and finite-row Ramanujan leakage. A raw row beyond R should not be called centered until the first term has been removed.

3 Recentered Type-I closure at the square-root wall

The row identity (27) separates the part seen by Bombieri–Vinogradov from two elementary terms. This permits the row modulus to extend beyond the spectral cutoff R.

For $F \in C_c^1((0,\infty))$, write

$$\mathcal{R}_{m,h;R}(X;F) = \sum_{j\geq 1} r_R(mj+h)F(mj/X). \tag{29}$$

Theorem 3.1 (Recentered full Type-I row ball). *Fix $h \neq 0$, $0 < \delta < 1/2$, and $F \in C_c^1((0,\infty))$. Put $R = \lfloor X^{1/2-\delta} \rfloor$. For every $A > 0$ there is $B = B(A,h,F) > 0$ such that, whenever*

$$M \leq \frac{X^{1/2}}{(\log X)^B}, \tag{30}$$

one has uniformly over all $|\alpha_m| \leq 1$

$$\left|\sum_{m\leq M} \alpha_m \left(\mathcal{R}_{m,h;R}(X;F) - \Delta_{h,R}(m)\frac{X}{m}I_0(F)\right)\right| \ll_{A,h,F,\delta} X(\log X)^{-A} + MR(\log(2R))^2. \tag{31}$$

In particular, with M as in (11), the right side is $O_A(X(\log X)^{-A})$ for every $A > 0$.

Proof. Suppose first that $(m,h) = 1$. Smooth maximal Bombieri–Vinogradov gives

$$\sum_{\substack{m\leq M\\(m,h)=1}} |\mathcal{E}_{m,h}(X;F)| \ll_{A,h,F} X(\log X)^{-A} \tag{32}$$

after increasing the requested logarithmic exponent. This is the usual fixed residue $h \pmod m$ application of the theorem [1, 7, 10].

If $(m,h) > 1$, a nonzero value $\Lambda(mj+h)$ forces $mj+h$ to be a power of one of the finitely many primes dividing h. On the fixed multiplicative support of F, there are $O_{h,F}(1)$ relevant powers for each such prime. Summing over m counts divisors of the finitely many integers $p^a - h$, and therefore contributes $O_{h,F,\varepsilon}(X^\varepsilon)$. The local density $\rho_h(m)$ is zero on these rows, so this is exactly the required treatment of (28); no reduced residue-class main term is used.

Insert (27). The prime errors are bounded by (32) and the preceding prime-power argument. The model leakage is uniform in m and sums to $O_F(MR(\log(2R))^2)$. Taking the supremum over the masks is harmless after the absolute sum over row errors. This proves (31).

For (11), one has $M \le X^{1/2}(\log X)^{-B}$ for every fixed B and all sufficiently large X. Also

$$MR(\log(2R))^2 \ll X^{1-\delta}\exp(-\sqrt{\log X})(\log X)^2,$$

which is $O_A(X(\log X)^{-A})$ for every fixed A. Here and below this assertion is pointwise in each fixed A: the Bombieri–Vinogradov exponent and the threshold in X may depend on A. No estimate uniform in an unbounded parameter A is intended. □

The drift itself remains negligible for the logarithmic coefficient vectors produced by Vaughan's identity.

Corollary 3.2 (Bounded-log uncentered rows). *Under the hypotheses of theorem 3.1, take M from (11). Uniformly for $|a_m| \le \log X$,*

$$\sum_{m\le M} a_m \mathcal{R}_{m,h;R}(X;F) \ll_{A,h,F,\delta} X(\log X)^{-A} + \frac{X}{R}(\log X)^3 \log\log X. \tag{33}$$

Consequently the left side is $O_A(X(\log X)^{-A})$ for every $A > 0$.

Proof. Divide a_m by $\log X$ and apply theorem 3.1 with one additional logarithmic saving. By lemma 2.2, the removed means contribute at most

$$X\log X \sum_{m\le M} \frac{|\Delta_{h,R}(m)|}{m} \ll_h \frac{X}{R}(\log X)^2 \log\log X.$$

We recorded the slightly looser exponent in (33). Since $R = X^{1/2-\delta+o(1)}$ and $\delta < 1/2$, this power-saving term is $O_A(X(\log X)^{-A})$ for every fixed A. □

Remark 3.3 (What has been extended). The finite model is exactly calibrated row by row only for $m \le R$. The theorem does not change that fact. It proves instead that the explicit sum of missing local factors is harmless throughout the classical averaged progression range. This is why the analytic row wall can move while the spectral cutoff remains fixed.

We also need one elementary logarithmic row. It is included for clarity rather than hidden in the Vaughan proof.

Lemma 3.4 (The modulus-one logarithmic row). *For every $A > 0$,*

$$\sum_{j\ge1} (\log j) r_R(j+h) W(j/X) \ll_{A,h,W,\delta} X(\log X)^{-A}. \tag{34}$$

Proof. The prime number theorem with its classical zero-free-region error and partial summation evaluate the $\Lambda(j+h)$ sum with continuous main term $\int(\log t)W(t/X)\,dt$. The complete-period mean of $\Lambda_R(j+h)$ is 1, and the logarithmic version of lemma 2.3 has error $O_W(R(\log(2XR))^3)$. The two continuous main terms agree. Since $R = X^{1/2-\delta+o(1)}$, both remaining errors have arbitrary logarithmic saving relative to X. □

4 Asymmetric Vaughan normal forms

For integers $U, V \ge 1$, Vaughan's identity in the sign convention used here is

$$\sum_t \Lambda(t)F(t) = \sum_{t\le U}\Lambda(t)F(t) - \sum_{\substack{\ell\le U\\ d\le V}} \Lambda(\ell)\mu(d)\sum_{j\ge1}F(\ell dj)$$
$$+ \sum_{d\le V}\mu(d)\sum_{j\ge1}(\log j)F(dj) - \sum_{\substack{\ell>U\\ k>1}}\Lambda(\ell)\beta_V(k)F(\ell k), \tag{35}$$

where β_V is defined in (6). Since $\beta_V(k) = 0$ for $1 < k \le V$, the last term is supported on $k > V$. The identity is classical [5, 9].

The TPC-15 proof did not require $U = V$. Keeping the parameters free gives the following useful invariance.

Proposition 4.1 (Vaughan parameter invariance below the cutoff)**.** *Let $1 \le U, V$ satisfy $UV \le R$. Define*

$$\mathcal{H}_{h,R;U,V}(X) = \sum_{\substack{\ell > U \\ k > V}} \Lambda(\ell)\beta_V(k) r_R(\ell k + h) W(\ell k/X). \tag{36}$$

Then for every $A > 0$,

$$\mathcal{P}_h(X; W) = \mathfrak{S}(h) X I_0(W) - \mathcal{H}_{h,R;U,V}(X) + O_{A,h,W,\delta}(X(\log X)^{-A}). \tag{37}$$

Consequently any two admissible parameter pairs give hard packets that differ by $O_A(X(\log X)^{-A})$.

Proof. TPC-15 proves the model cross-correlation

$$\sum_t \Lambda(t)\Lambda_R(t+h) W(t/X) = \mathfrak{S}(h) X I_0(W) + O_A(X(\log X)^{-A}). \tag{38}$$

Apply (35) to $F(t) = r_R(t+h)W(t/X)$. The support of W makes the first packet zero because $U \le R = o(X)$. In the second packet the row modulus is $m = \ell d \le UV \le R$, and its total coefficient at a fixed m is at most $\sum_{\ell | m} \Lambda(\ell) = \log m$. The third packet has $d \le V \le R$ and the standard logarithmic row multiplier. The full bounded Type-I row theorem of TPC-15 controls both packets with arbitrary logarithmic saving. The final term is $-\mathcal{H}_{h,R;U,V}$. Combining with (38) proves the proposition. □

At the edge of proposition 4.1, the choice $U = R, V = 1$ gives $\beta_1(k) = 1$. The recentered theorem permits U to move much farther.

Theorem 4.2 (Square-root Vaughan normal form)**.** *Let M be given by (11) and define*

$$\mathcal{H}^{\sharp}_{h,R;M}(X) = \sum_{\substack{\ell > M \\ k > 1}} \Lambda(\ell) r_R(\ell k + h) W(\ell k/X). \tag{39}$$

Then, for every $A > 0$,

$$\mathcal{P}_h(X; W) = \mathfrak{S}(h) X I_0(W) - \mathcal{H}^{\sharp}_{h,R;M}(X) + O_{A,h,W,\delta}(X(\log X)^{-A}). \tag{40}$$

In particular, when h is an admissible even shift and $I_0(W) \neq 0$,

$$\mathcal{P}_h(X; W) \sim \mathfrak{S}(h) X I_0(W) \quad \Longleftrightarrow \quad \mathcal{H}^{\sharp}_{h,R;M}(X) = o(X). \tag{41}$$

Proof. Again split the target as $\Lambda(t+h) = \Lambda_R(t+h) + r_R(t+h)$ and use (38). Apply (35) to the residual test function with $U = M, V = 1$. The first packet vanishes by support. The second packet is

$$-\sum_{\ell \le M} \Lambda(\ell) \sum_j r_R(\ell j + h) W(\ell j/X),$$

which is $O_A(X(\log X)^{-A})$ by corollary 3.2. The third is precisely the modulus-one logarithmic row in lemma 3.4. Finally, $\beta_1(k) = 1$, so the last packet is $-\mathcal{H}^{\sharp}_{h,R;M}$. This proves (40). The equivalence follows because the displayed remainder is $o(X)$ and the stated main coefficient is nonzero. □

Corollary 4.3 (Equivalence with the TPC-15 packet)**.** *With $U = V = \lfloor R^{1/2} \rfloor$,*

$$\mathcal{H}^{\mathrm{HB}}_{h,R;U,V}(X) = \mathcal{H}^{\sharp}_{h,R;M}(X) + O_A(X(\log X)^{-A}). \tag{42}$$

This equality is an equality of total signed packets. It does not match individual factor pairs, and it must not be read as a positivity-preserving transport from one dyadic scale to another.

5 Dyadic geometry and the surviving right wing

Choose a nonnegative $\psi \in C_c^\infty([1/2,2])$ satisfying

$$\sum_{j\in\mathbb{Z}} \psi(x/2^j) = 1 \qquad (x>0). \tag{43}$$

For dyadic L, K, put

$$\mathcal{H}^\sharp_{L,K}(h;X) = \sum_{\substack{\ell>M \\ k>1}} \Lambda(\ell) r_R(\ell k+h) W(\ell k/X)\psi(\ell/L)\psi(k/K). \tag{44}$$

Proposition 5.1 (Exact dyadic corridor)**.** *One has the finite exact decomposition*

$$\mathcal{H}^\sharp_{h,R;M}(X) = \sum_{L,K} \mathcal{H}^\sharp_{L,K}(h;X). \tag{45}$$

If $\operatorname{supp} W \subset [a,b] \subset (0,\infty)$*, every nonzero block satisfies*

$$\frac{aX}{4} \le LK \le 4bX, \qquad L > M/2, \qquad K \ll_W X/M. \tag{46}$$

There are $O_W(\log X)$ relevant pairs (L,K).

Proof. Insert (43) in both variables. Compact support of W and the lower cutoffs make the sum finite. If the two dyadic factors are nonzero, then $\ell \asymp L$, $k \asymp K$, and $\ell k \in [aX, bX]$, giving the product bounds. For each L, the product condition permits only $O_W(1)$ dyadic values of K. There are $O(\log X)$ possible L between $M/2$ and $O_W(X)$. □

The normal form removes every physical term with $\ell \le M$. Consequently every dyadic block with $L \le M/2$ vanishes, while every surviving block satisfies $L > M/2$. Up to the fixed support constants of the dyadic partition, the closed left wing is therefore

$$L \ll X^{1/2}\exp(-\sqrt{\log X}). \tag{47}$$

The surviving set divides naturally into

$$\text{near-square-root core:} \qquad X^{1/2}\mathrm{e}^{-\sqrt{\log X}} \ll L, K \ll X^{1/2}\mathrm{e}^{\sqrt{\log X}}, \tag{48}$$

$$\text{right wing:} \qquad L \gg X^{1/2}\mathrm{e}^{\sqrt{\log X}}, \qquad K \ll X^{1/2}\mathrm{e}^{-\sqrt{\log X}}. \tag{49}$$

Here and below e^x denotes the ordinary exponential in prose-style scale displays; it is unrelated to the additive character $\mathrm{e}(t) = \exp(2\pi i t)$ used later.

Corollary 5.2 (Dyadic witness)**.** *If along a sequence X_j one has $|\mathcal{H}^\sharp_{h,R;M}(X_j)| \ge \eta X_j$ for some $\eta > 0$, then for every such X_j there is a relevant dyadic pair (L_j, K_j) with*

$$|\mathcal{H}^\sharp_{L,K}(h;X_j)| \gg_W \frac{\eta X_j}{\log X_j}. \tag{50}$$

Conversely, the uniform estimate

$$\max_{L,K} |\mathcal{H}^\sharp_{L,K}(h;X)| = o\left(\frac{X}{\log X}\right) \tag{51}$$

implies $\mathcal{H}^\sharp = o(X)$.

The first assertion is only a pigeonhole statement. It does not identify one fixed dyadic scale across all X, and the converse condition is sufficient rather than necessary because blocks can cancel.

5.1 A fixed quotient is already a two-form problem

It is tempting to call (49) easy because k is small. The coefficient $\Lambda(\ell)$, however, depends on the quotient in the row $k\ell+h$. Even a fixed slice contains an unresolved two-linear-form correlation.

Fix $k \geq 2$ and put $Y = X/k$. Define

$$\mathcal{C}_{k,h}(Y;W) = \sum_{\ell \geq 1} \Lambda(\ell)\Lambda(k\ell+h)W(\ell/Y), \tag{52}$$

$$\mathcal{E}_{k,h;R}(Y;W) = \sum_{\ell \geq 1} \Lambda(\ell) r_R(k\ell+h)W(\ell/Y). \tag{53}$$

Let

$$\mathfrak{S}(k,h) = \prod_p \frac{1-\nu_p(k,h)/p}{(1-1/p)^2}, \quad \nu_p(k,h) = \#\{a \pmod p : a(ka+h) \equiv 0 \pmod p\}. \tag{54}$$

For the finite model, put

$$S_q(k,h) = \sum_{\substack{a \pmod q \\ (a,q)=1}} c_q(ka+h), \qquad \mathfrak{S}_R(k,h) = \sum_{q \leq R} \frac{\mu(q)S_q(k,h)}{\varphi(q)^2}. \tag{55}$$

Proposition 5.3 (Fixed-k boundary)**.** *Assume that the two forms n and $kn+h$ are admissible. For fixed k, h and $R = X^{1/2-\delta}$,*

$$\sum_\ell \Lambda(\ell)\Lambda_R(k\ell+h)W(\ell/Y) = \mathfrak{S}(k,h)YI_0(W) + o_{k,h,W}(Y). \tag{56}$$

Consequently

$$\mathcal{C}_{k,h}(Y;W) \sim \mathfrak{S}(k,h)YI_0(W) \iff \mathcal{E}_{k,h;R}(Y;W) = o(Y), \tag{57}$$

provided $I_0(W) \neq 0$.

Proof. Admissibility implies $(k,h)=1$. Since k is fixed and $X = kY$,

$$R = (kY)^{1/2-\delta} + O(1) \leq \frac{Y^{1/2}}{(\log Y)^B} \tag{58}$$

for every fixed B and all sufficiently large Y.

Insert the divisor expansion (24). If $(d,k) > 1$, the congruence $d \mid k\ell+h$ has no solution. If $(d,k)=1$, it is the single class

$$\ell \equiv -h\overline{k} \pmod d, \tag{59}$$

which is reduced precisely when $(d,h)=1$. Maximal Bombieri–Vinogradov, (58), and (25) therefore give

$$\sum_\ell \Lambda(\ell)\Lambda_R(k\ell+h)W(\ell/Y) = YI_0(W) \sum_{\substack{d \leq R \\ (d,kh)=1}} \frac{\lambda'_R(d)}{\varphi(d)} + O_{A,k,h,W}(Y(\log Y)^{-A}). \tag{60}$$

Indeed, the coefficient costs only two additional logarithmic powers. If $(d,k)=1$ but $(d,h)>1$, a nonzero source weight forces $\ell = p^\nu$ for one of the finitely many $p \mid h$. Only $O_{h,W}(1)$ exponents lie on the support, and summing the divisor coefficients over divisors of $kp^\nu + h$ contributes $O_{k,h,W,\varepsilon}(Y^\varepsilon)$.

A finite rearrangement of the divisor and Ramanujan expansions gives

$$\sum_{\substack{d \le R \\ (d,kh)=1}} \frac{\lambda'_R(d)}{\varphi(d)} = \mathfrak{S}_R(k,h). \tag{61}$$

For completeness, this is also obtained by averaging $\Lambda_R(ka+h)$ over reduced residue classes a at each Ramanujan denominator.

For squarefree q, the summand in (55) is multiplicative. Its prime values are

$$\frac{\mu(p)S_p(k,h)}{\varphi(p)^2} = \begin{cases} -1/(p-1)^2, & p \nmid kh, \\ 1/(p-1), & p \mid k,\ p \nmid h, \\ 1/(p-1), & p \mid h,\ p \nmid k, \\ -1, & p \mid (k,h). \end{cases} \tag{62}$$

Thus one plus the local value is exactly $(1-\nu_p(k,h)/p)/(1-1/p)^2$. The last case is absent under admissibility. The infinite Ramanujan series is absolutely convergent, equals (54), and satisfies

$$\mathfrak{S}_R(k,h) = \mathfrak{S}(k,h) + O_{k,h}\left(\frac{(\log(2R))^2}{R}\right). \tag{63}$$

Combining (60)– (63) proves (56). Finally, split $\Lambda(k\ell+h) = \Lambda_R(k\ell+h) + r_R(k\ell+h)$ to obtain (57). □

For example, when $h = 2$ and k is odd, the forms n and $kn+2$ are admissible. Their conjectural simultaneous primality is a special case of the Hardy–Littlewood/Dickson prime-tuples conjecture [4]. The proposition does not say that a bound for the total right wing must solve every fixed-k slice separately; cross-k cancellation remains possible. It does prove that the right wing cannot be discarded merely by relabeling it Type I.

6 Three-scale peeling and a near-$11/21$ factorable wedge

The square-root normal form is best for locating the source-row wall. To use factorable-modulus distribution theorems, we return to the symmetric TPC-15 packet with $U = V = \lfloor R^{1/2} \rfloor$ and open (6). Writing $k = dj$ reveals the progression modulus

$$q = \ell d. \tag{64}$$

Thus the relevant geometry has three factor scales (L, K, D), with the quotient length $J \asymp K/D$.

6.1 The exact invisible wedge

Define

$$\gamma_{\ell;R,V}(k) = \sum_{\substack{d \mid k \\ d \le V \\ \ell d > R}} \mu(d). \tag{65}$$

Proposition 6.1 (Hyperbolic Type-I peeling)**.** *For every $A > 0$,*

$$\begin{aligned} \mathcal{H}^{\mathrm{HB}}_{h,R;U,V}(X) = &\sum_{\substack{\ell > U \\ k > V}} \Lambda(\ell)\gamma_{\ell;R,V}(k) r_R(\ell k + h) W(\ell k/X) \\ &+ O_{A,h,W,\delta}(X(\log X)^{-A}). \end{aligned} \tag{66}$$

In particular, if $U < \ell \le R$, only divisors $d > R/\ell$ survive; if $\ell > R$, the coefficient is the full $\beta_V(k)$.

Proof. The difference between (7) and the main term in (66) is

$$\sum_{\substack{\ell>U,\ d\le V\\ \ell d\le R}} \Lambda(\ell)\mu(d)\sum_{j\ge 1} r_R(\ell dj+h)W(\ell dj/X), \tag{67}$$

for all sufficiently large X. Indeed, on this support $\ell \le R$ and $\ell dj \asymp_W X$, so $k = dj \gg_W X/R \gg V$; the original restriction $k > V$ is automatic.

Group (67) by $m = \ell d \le R$. Its row coefficient is

$$A_m = \sum_{\substack{\ell d=m\\ \ell>U,\ d\le V}} \Lambda(\ell)\mu(d), \qquad |A_m| \le \sum_{\ell\mid m} \Lambda(\ell) = \log m.$$

The full bounded Type-I row theorem of TPC-15, with one extra logarithmic saving, makes (67) $O_A(X(\log X)^{-A})$. □

The curve $\ell d = R$ is an exact calibration boundary for this decomposition. It is not an absolute boundary for all analytic methods; factorable-modulus theorems can enter part of the exterior.

6.2 An imported fixed-residue distribution theorem

We use the following consequence of Maynard [6, Theorem 1.1]. It is stated in a form adapted to smooth von Mangoldt sums.

Theorem 6.2 (Maynard, fixed residue and factorable moduli)**.** *Fix $a \ne 0$, $\eta > 0$, and $A > 0$. Suppose*

$$Q_1Q_2^2 \le X^{1-\eta}, \qquad Q_1^{12}Q_2^7 \le X^{4-\eta}, \qquad Q_1^{20}Q_2^{19} \le X^{10-\eta}. \tag{68}$$

Let $G_{q_1,q_2} \in C_c^1((0,\infty))$ range over a family with a common compact support and a common C^1 bound. Then

$$\sum_{\substack{q_1\le Q_1,\ q_2\le Q_2\\ (q_1q_2,a)=1}} \left| \sum_{\substack{n\ge 1\\ n\equiv a \pmod{q_1q_2}}} \Lambda(n)G_{q_1,q_2}((n-a)/X) - \frac{X}{\varphi(q_1q_2)} I_0(G_{q_1,q_2}) \right| \ll_{a,\eta,A} X(\log X)^{-A}. \tag{69}$$

Derivation of the stated smooth form. Maynard's theorem is stated for

$$E(t;q,a) = \pi(t;q,a) - \frac{\pi(t)}{\varphi(q)}.$$

Stieltjes partial summation, followed by the triangle inequality and Fubini, bounds the sum of the prime-only weighted errors by a constant times

$$(\log X)\sup_{t\asymp X}\sum_{q_1,q_2} |E(t;q_1q_2,a)| + \frac{\log X}{X}\int_{t\asymp X}\sum_{q_1,q_2} |E(t;q_1q_2,a)|\,dt.$$

The common support and C^1 bounds make this estimate uniform even though G_{q_1,q_2} depends on the ordered modulus pair. Apply Maynard [6, Theorem 1.1] pointwise in $t \asymp X$, with its parameter chosen smaller than $\eta/100$ and with two additional logarithmic powers requested. This gives $O_A(X(\log X)^{-A})$.

The global prime number theorem, uniformly for the same bounded weight family, gives

$$\sum_p (\log p)G_{q_1,q_2}((p-a)/X) = XI_0(G_{q_1,q_2}) + O\left(X\exp(-c\sqrt{\log X})\right).$$

Its aggregate error is negligible because

$$\sum_{q_1 \le Q_1, q_2 \le Q_2} \frac{1}{\varphi(q_1 q_2)} \le \left(\sum_{q_1 \le Q_1} \frac{1}{\varphi(q_1)} \right) \left(\sum_{q_2 \le Q_2} \frac{1}{\varphi(q_2)} \right) \ll (\log X)^C.$$

Finally, restore target prime powers. If $p^\nu \asymp X$, $\nu \ge 2$, and $p^\nu \equiv a \pmod{q_1 q_2}$, then $q_1 q_2 \mid p^\nu - a$. Consequently their total over the ordered pairs is

$$\ll \sum_{\substack{p^\nu \asymp X \\ \nu \ge 2}} (\log p) \tau_3(|p^\nu - a|) \ll_\varepsilon X^{1/2+\varepsilon}.$$

This proves (69). The absolute values in Maynard's theorem are what permit dyadic restrictions and bounded outer coefficients. No new distribution estimate is proved here. □

Let $\Phi_1, \Phi_2, \Phi_3 \in C_c^\infty([1,2])$. Define the smooth D-resolved hard subpacket

$$\mathcal{H}_{h,R}(L,K,D) = \sum_{\ell,d,j \ge 1} \Lambda(\ell)\mu(d)\Phi_1(\ell/L)\Phi_2(d/D)\Phi_3(dj/K) \times r_R(\ell dj + h) W(\ell dj/X). \tag{70}$$

Theorem 6.3 (A factorable exterior wedge)**.** *Fix $h \ne 0$, $0 < \delta < 1/2$, and $\eta > 0$. Suppose the scales satisfy*

$$LK \asymp_W X, \qquad 2D < R < L/2, \qquad 2D \le V, \qquad K > 2V, \qquad L > 2U, \tag{71}$$

$$DL^2 \le X^{1-\eta}, \qquad D^{12}L^7 \le X^{4-\eta}, \qquad D^{20}L^{19} \le X^{10-\eta}, \tag{72}$$

$$LDR \le X^{1-\eta}. \tag{73}$$

Then for every $A > 0$,

$$\mathcal{H}_{h,R}(L,K,D) \ll_{A,h,W,\Phi_1,\Phi_2,\Phi_3,\delta,\eta} X(\log X)^{-A}. \tag{74}$$

Proof. The contribution of source prime powers $\ell = p^a$, $a \ge 2$, is removed first. The divisor expansion gives $|r_R(n)| \ll_\varepsilon X^\varepsilon$ on the support in question. Hence their entire subpacket is

$$\ll_\varepsilon L^{1/2} D \frac{K}{D} X^\varepsilon \ll XL^{-1/2+\varepsilon}, \tag{75}$$

because $LK \asymp X$. This has a fixed power saving since $L > 2U$. We may therefore assume below that ℓ is prime. In particular, $\ell > L > 2R > |h|$ for all sufficiently large X.

Fix $\ell \asymp L$ and $d \asymp D$, and set $q = \ell d$. The inner sum is a smooth progression row of modulus q, with a family of weights whose supports and C^1 norms are uniform because $LK \asymp X$. Equation (27) decomposes it into prime error, cutoff drift, and model leakage.

On rows with $(\ell d, h) = 1$, apply theorem 6.2 with $Q_1 = 2D, Q_2 = 2L$. The powers of 2 are absorbed by the strict $X^{-\eta}$ margins. The theorem sums absolute errors, so the restrictions $d \asymp D$, $\ell \asymp L$ and the weights $|\mu(d)|$, $\Lambda(\ell) \ll \log X$ cost only a change in the requested logarithmic saving. This bounds the total prime-progression error by $O_A(X(\log X)^{-A})$.

If $(\ell d, h) > 1$, the common prime lies in d, because the remaining ℓ is prime and exceeds $|h|$. If $p \mid (d,h)$, then $\ell dj + h$ is divisible by p; a nonzero von Mangoldt value therefore forces $\ell dj + h = p^\nu$. For each of the finitely many $p \mid h$, only $O_{h,W}(1)$ exponents lie on the X scale, and the number of ordered factorizations $\ell d \mid p^\nu - h$ is $O_\varepsilon(X^\varepsilon)$. Thus the total prime part of the nonreduced rows is negligible. Their complete model mean is zero: all divisors of d lie below R, and their local Euler factor contains $1 - 1$ at a prime dividing (d,h).

The finite-row model leakage is $O(R(\log(2R))^2)$ per pair (ℓ, d). After the outer logarithmic weight and the $O(LD)$ pairs, its total is

$$\ll LDR(\log X)^3, \tag{76}$$

which has a fixed power saving by (73).

It remains to treat the drift. Since $d < R < \ell$ and $\mu(d) \neq 0$, every divisor of ℓd below R is a divisor of d. Therefore

$$\Delta_{h,R}(\ell d) = \begin{cases} \dfrac{d}{\varphi(d)(\ell - 1)}, & (d,h) = 1, \\ 0, & (d,h) > 1. \end{cases} \tag{77}$$

Indeed, the model mean is $d/\varphi(d)$ and the prime mean is $\ell d/(\varphi(\ell)\varphi(d))$ in the first case; both are zero in the second. The continuous drift contribution is consequently

$$\begin{aligned} &\ll X \sum_{\ell \asymp L} \frac{\Lambda(\ell)}{\ell(\ell-1)} \sum_{d \asymp D} \frac{\mu(d)^2}{\varphi(d)} \\ &\ll \frac{X}{L}(\log(2D))^2. \end{aligned} \tag{78}$$

This also has a fixed power saving. Combining the three pieces proves (74). □

Corollary 6.4 (A packet near the 11/21 factorable boundary)**.** *Fix* $0 < \sigma < 1/100$ *and*

$$\frac{1}{42} + 3\sigma < \delta < \frac{1}{4}. \tag{79}$$

Take

$$D = X^{1/21-\sigma}, \qquad L = X^{10/21-\sigma}, \qquad K = X/L = X^{11/21+\sigma}. \tag{80}$$

Then, taking $\eta = 2\sigma$ *in theorem 6.3, the packet* (70) *is* $O_A(X(\log X)^{-A})$ *for every* $A > 0$.

Proof. The three Maynard monomials have exponents

$$1 - 3\sigma, \qquad \frac{82}{21} - 19\sigma < 4, \qquad 10 - 39\sigma.$$

Moreover,

$$LDR = X^{1+1/42-2\sigma-\delta} \leq X^{1-5\sigma}$$

by (79). The same range gives $R < L$; the upper bound $\delta < 1/4$ ensures $2D < V$ for large X. All remaining support conditions in theorem 6.3 are immediate. □

Both L and K exceed R in this corollary, so the packet lies inside the central (L, K) corridor of the original TPC-15 representation. The result is nevertheless only one D-resolved slice. Summing over all D, or removing the Maynard factorization inequalities, is not justified by the theorem.

7 Residual energy and an exact translation matrix coefficient

The finite model matches the local mean and the fixed-shift main term, but it does not approximate Λ in mean square. We quantify that fact before formulating the remaining gate spectrally.

Put

$$A_R = \sum_{q \leq R} \frac{\mu(q)^2}{\varphi(q)}. \tag{81}$$

Lemma 7.1 (The diagonal model mass)**.** *One has*

$$A_R = \log R + C_0 + o(1) \tag{82}$$

for an absolute constant C_0.

Proof. The Dirichlet series of $\mu(n)^2/\varphi(n)$ is

$$\prod_p \left(1 + \frac{1}{(p-1)p^s}\right) = \zeta(s+1)G(s),$$

where the local factors of G are

$$G_p(s) = 1 + \frac{p^{-s} - p^{-2s}}{p(p-1)}.$$

Thus G is absolutely analytic for $\Re s > -1/2$, and

$$G(0) = \prod_p \left(1 + \frac{1}{p-1}\right)(1 - 1/p) = 1.$$

A standard Perron–Selberg–Delange argument gives (82). $\square$

Theorem 7.2 (Exact leading residual energy)**.** *Let $F \in C_c^\infty((0,\infty);\mathbb{R})$ be nonnegative. For fixed $0 < \delta < 1/2$ and $R = \lfloor X^{1/2-\delta} \rfloor$,*

$$\sum_n r_R(n)^2 F(n/X) = X \log(X/R) I_0(F) + O_{F,\delta}(X). \tag{83}$$

Equivalently,

$$\sum_n r_R(n)^2 F(n/X) = \left(\frac{1}{2} + \delta + o(1)\right) X \log X \, I_0(F). \tag{84}$$

Moreover,

$$\sum_n r_R(n)\Lambda_R(n)F(n/X) = O_{F,\delta}(X), \tag{85}$$

so the model and residual are asymptotically orthogonal at the $X \log X$ scale.

Proof. The prime number theorem and partial summation give

$$\sum_n \Lambda(n)^2 F(n/X) = X \log X \, I_0(F) + O_F(X). \tag{86}$$

Prime powers of exponent at least two contribute $o(X)$.

If $p \asymp_F X$ is prime, then $p > R$ and $(p,q) = 1$ for every $q \leq R$. Hence $c_q(p) = \mu(q)$ and

$$\Lambda_R(p) = A_R. \tag{87}$$

The higher prime powers are again negligible using the divisor expansion and the divisor bound; explicitly, for a fixed support,

$$\sum_{\substack{p^\nu \asymp_F X \\ \nu \geq 2}} \Lambda(p^\nu)|\Lambda_R(p^\nu)| \ll_F X^{1/2}(\log X)^C = o(X)$$

for some absolute C. Therefore

$$\sum_n \Lambda(n)\Lambda_R(n)F(n/X) = X A_R I_0(F) + O_F(X). \tag{88}$$

Finally, insert the rational Fourier expansion of Λ_R twice. The zero-frequency pairs give

$$XI_0(F)\sum_{q\le R}\frac{\mu(q)^2c_q(0)}{\varphi(q)^2}=XA_RI_0(F).$$

All nonzero frequency differences are at least R^{-2} apart. The off-diagonal absolute bound furnished by Poisson summation is, for every $B>0$,

$$\ll_{B,F} XR^2\left(1+\frac{X}{R^2}\right)^{-B}.$$

Since $R^2\le X^{1-2\delta}$, choosing B in terms of δ gives arbitrary logarithmic saving, so

$$\sum_n \Lambda_R(n)^2F(n/X)=XA_RI_0(F)+O_{F,\delta}(X(\log X)^{-10}). \tag{89}$$

Expanding $(\Lambda-\Lambda_R)^2$ and using lemma 7.1 proves (83). Subtracting (89) from (88) gives (85). $\square$

7.1 The hard packet as a translation correlation

Define the square-root Vaughan coefficient on the integer line by

$$a_X(n)=W(n/X)\sum_{\substack{\ell k=n\\ \ell>M\\ k>1}}\Lambda(\ell). \tag{90}$$

Choose $\chi\in C_c^\infty((0,\infty))$ equal to 1 on a fixed neighborhood of $\operatorname{supp} W$, and put

$$g_X(n)=r_R(n)\chi(n/X). \tag{91}$$

Let $(T^hg)(n)=g(n+h)$. We use the convention

$$\langle f,g\rangle_{\ell^2(\mathbb{Z})}=\sum_{n\in\mathbb{Z}}f(n)\overline{g(n)}. \tag{92}$$

Proposition 7.3 (Translation and cross-spectrum identities)**.** *For all sufficiently large X,*

$$\mathcal{H}^\sharp_{h,R;M}(X)=\langle a_X,T^hg_X\rangle_{\ell^2(\mathbb{Z})}. \tag{93}$$

With Fourier convention

$$\widehat{f}(\theta)=\sum_{n\in\mathbb{Z}}f(n)\mathrm{e}(-n\theta), \tag{94}$$

one has

$$\mathcal{H}^\sharp_{h,R;M}(X)=\int_{\mathbb{T}}\widehat{a}_X(\theta)\overline{\widehat{g}_X(\theta)}\mathrm{e}(-h\theta)\,d\theta. \tag{95}$$

Proof. Group (39) by $n=\ell k$. Since h is fixed and $\chi=1$ near $\operatorname{supp} W$, one has $g_X(n+h)=r_R(n+h)$ on the support for all sufficiently large X, proving (93). Parseval and $\widehat{T^hg}(\theta)=\mathrm{e}(h\theta)\widehat{g}(\theta)$ give (95). $\square$

The spectral language here is literal but elementary. The eigenphases belong to translation on $\mathbb{Z}$ and are indexed by $\theta\in\mathbb{T}$; h is a time lag. Neither primes nor zeta zeros are asserted to be eigenvalues.

7.2 A quantitative angular corridor

The coefficient has sharp enough norm bounds for a normalized coherence to be meaningful.

Lemma 7.4 (Coefficient norm ledger). *If $W \not\equiv 0$, then*

$$c_W X \le \|a_X\|_2^2 \le C_W X(\log X)^2. \tag{96}$$

Also,

$$\|g_X\|_2^2 = \left(\frac{1}{2} + \delta + o(1)\right) X \log X\, I_0(\chi^2). \tag{97}$$

Proof. The divisor identity $\sum_{d|n} \Lambda(d) = \log n$ gives

$$|a_X(n)| \le \|W\|_\infty \log n,$$

which proves the upper bound. For the lower bound, choose two compact prime intervals of scale $\sqrt{X}$ whose products lie where $|W|$ is bounded below. For distinct primes p, q in those intervals, both exceed M and

$$a_X(pq) = W(pq/X)(\log p + \log q) = W(pq/X)\log(pq).$$

The prime number theorem supplies $\gg_W X/(\log X)^2$ distinct such semiprimes, proving the lower bound. Equation (97) is theorem 7.2 with $F = \chi^2$. $\square$

Define the normalized fixed-shift coherence

$$\kappa_{h,R}(X) = \frac{\mathcal{H}^\sharp_{h,R;M}(X)}{\|a_X\|_2\|g_X\|_2}. \tag{98}$$

Then $|\kappa_{h,R}(X)| \le 1$ by Cauchy–Schwarz and

$$X\sqrt{\log X} \ll_{W,\chi,\delta} \|a_X\|_2\|g_X\|_2 \ll_{W,\chi,\delta} X(\log X)^{3/2}. \tag{99}$$

Corollary 7.5 (Angular sufficient and necessary scales). *The estimate*

$$\kappa_{h,R}(X) = o((\log X)^{-3/2}) \tag{100}$$

is sufficient for $\mathcal{H}^\sharp = o(X)$. Conversely, $\mathcal{H}^\sharp = o(X)$ implies

$$\kappa_{h,R}(X) = o((\log X)^{-1/2}). \tag{101}$$

The gap between the two logarithmic exponents comes only from the present upper and lower bounds for $\|a_X\|_2$. More importantly, the theorem shows why energy decay cannot solve the gate: the residual energy in (84) is of order $X \log X$. What is missing is a small angle with one specified arithmetic direction.

8 Phase-averaged control and the distinguished zero phase

The angular formulation in section 7 is at the integer-translation level. There is a complementary factor-coordinate test: modulate the source and quotient variables independently. Define

$$\mathcal{H}^\sharp(\alpha, \beta) = \sum_{\substack{\ell > M \\ k > 1}} \Lambda(\ell) r_R(\ell k + h) W(\ell k/X) \mathrm{e}(\alpha\ell + \beta k). \tag{102}$$

The original packet is the distinguished point $\mathcal{H}^\sharp(0, 0)$.

For the dyadic estimates below, define

$$\mathcal{H}^\sharp_{L,K}(\alpha,\beta) = \sum_{\substack{\ell>M\\ k>1}} \Lambda(\ell) r_R(\ell k+h) W(\ell k/X)\psi(\ell/L)\psi(k/K)\mathrm{e}(\alpha\ell+\beta k). \tag{103}$$

We use the elementary divisor bound

$$|r_R(n)| \ll_{\varepsilon,h,W,\delta} X^\varepsilon \qquad (n-h \asymp_W X). \tag{104}$$

Indeed, (24), the pointwise estimate $|\lambda'_R(d)| \ll (d/\varphi(d))\log(2R/d)$, and the divisor bound give $|\Lambda_R(n)| \ll_\varepsilon X^\varepsilon$, while $\Lambda(n) \le \log n$.

Theorem 8.1 (Factor-phase mean squares)**.** *For every $\varepsilon > 0$,*

$$\int_{\mathbb{T}} |\mathcal{H}^\sharp(\alpha,0)|^2\,d\alpha \ll_{\varepsilon,h,W,\delta} \frac{X^{2+\varepsilon}}{M}, \tag{105}$$

$$\int_{\mathbb{T}}\int_{\mathbb{T}} |\mathcal{H}^\sharp(\alpha,\beta)|^2\,d\alpha\,d\beta \ll_{\varepsilon,h,W,\delta} X^{1+\varepsilon}. \tag{106}$$

For a dyadic block with $LK \asymp_W X$,

$$\int_{\mathbb{T}} |\mathcal{H}^\sharp_{L,K}(\alpha,0)|^2\,d\alpha \ll_\varepsilon \frac{X^{2+\varepsilon}}{L}, \tag{107}$$

$$\int_{\mathbb{T}} |\mathcal{H}^\sharp_{L,K}(0,\beta)|^2\,d\beta \ll_\varepsilon \frac{X^{2+\varepsilon}}{K}, \tag{108}$$

$$\int_{\mathbb{T}}\int_{\mathbb{T}} |\mathcal{H}^\sharp_{L,K}(\alpha,\beta)|^2\,d\alpha\,d\beta \ll_\varepsilon X^{1+\varepsilon}. \tag{109}$$

Proof. Parseval in α gives

$$\int_{\mathbb{T}} |\mathcal{H}^\sharp(\alpha,0)|^2 d\alpha = \sum_{\ell>M} \Lambda(\ell)^2 \left|\sum_{k>1} r_R(\ell k+h)W(\ell k/X)\right|^2. \tag{110}$$

For each ℓ, the inner sum has $O_W(X/\ell)$ terms. Apply (104), square, and use

$$\sum_{\ell>M} \frac{\Lambda(\ell)^2}{\ell^2} \ll \frac{\log(2M)}{M}.$$

The logarithm is absorbed by X^ε, proving (105).

Double Parseval gives the exact diagonal sum

$$\int_{\mathbb{T}}\int_{\mathbb{T}} |\mathcal{H}^\sharp(\alpha,\beta)|^2\,d\alpha\,d\beta = \sum_{\ell>M,k>1} \Lambda(\ell)^2 |r_R(\ell k+h)|^2 |W(\ell k/X)|^2.$$

The pointwise residual bound and $\sum_{\ell\le y}\Lambda(\ell)^2 \ll y\log(2y)$ make this $O_\varepsilon(X^{1+\varepsilon})$. Restricting the same arguments to $\ell \asymp L, k \asymp K$ proves equations (107) and (109). For (108), Parseval in β and Cauchy–Schwarz in ℓ give

$$\sum_{k\asymp K}\left|\sum_{\ell\asymp L} \Lambda(\ell) r_R(\ell k+h) W(\ell k/X)\psi(\ell/L)\right|^2 \ll_\varepsilon KL^2X^\varepsilon.$$

Here we used $\sum_{\ell\asymp L}\Lambda(\ell)^2 \ll L\log(2L)$ and absorbed the logarithm into X^ε. Since $LK \asymp X$, this is $O_\varepsilon(X^{2+\varepsilon}/K)$. □

Since $M = X^{1/2-o(1)}$, the root mean square in (105) is $X^{3/4+o(1)}$, and the double-phase root mean square is $X^{1/2+o(1)}$. More quantitatively, for every fixed $\eta > 0$, Chebyshev's inequality shows at each X that the set of row phases with $|\mathcal{H}^\sharp(\alpha, 0)| > X^{3/4+\eta}$ has measure $O_\eta(X^{-2\eta+o(1)})$; the analogous double-phase exceptional measure at threshold $X^{1/2+\eta}$ obeys the same bound. Along a sufficiently sparse dyadic subsequence, Borel–Cantelli gives the corresponding almost-everywhere statements. For independent Rademacher signs, the expectations of the squared magnitudes equal the matching diagonal sums in the Parseval calculations.

Remark 8.2 (Why the theorem does not evaluate the gate). The point $(\alpha, \beta) = (0, 0)$ has Haar measure zero. An L^2 estimate over phases can coexist with a spike at that point. In the original arithmetic problem all rows are coherently unmodulated, so no phase average is available. The theorem identifies a phase-averaged mask family with power cancellation; it does not transfer that cancellation to the fixed all-positive direction.

8.1 Operator norms and actual arithmetic directions

For a dyadic block define the residual matrix

$$\mathsf{R}^{(h,R)}_{L,K}(\ell, k) = r_R(\ell k + h) W(\ell k/X)\psi(\ell/L)\psi(k/K). \tag{111}$$

With $u_\ell = \Lambda(\ell)$ and, in the original TPC-15 coordinates, $v_k = \beta_V(k)$, the block is $u^*\mathsf{R}v$. Standard divisor moments give

$$\|u\|_2^2 \ll L\log(2L), \qquad \|v\|_2^2 \ll K(\log(2K))^3. \tag{112}$$

Thus a uniform operator estimate

$$\|\mathsf{R}^{(h,R)}_{L,K}\|_{2\to 2} = o\left(\frac{\sqrt{X}}{(\log X)^2}\right) \tag{113}$$

would make that block $o(X)$.

This is a sufficient condition for the actual arithmetic vectors. It is necessary and sufficient only for the full pair of ℓ^2 mask balls, because

$$\sup_{\|x\|_2 \le A,\ \|y\|_2 \le B} |x^*\mathsf{R}y| = AB\|\mathsf{R}\|_{2\to 2}. \tag{114}$$

The prime-pair problem asks for one specified coefficient $u^*\mathsf{R}v$, not the largest singular direction. Replacing the directional problem by (113) would therefore ask for a stronger theorem than is logically necessary.

The available coarse Hilbert–Schmidt ledger is also not enough. From (104) one obtains the coarse bound

$$\|\mathsf{R}^{(h,R)}_{L,K}\|_{\mathrm{HS}}^2 \ll_\varepsilon X^{1+\varepsilon}. \tag{115}$$

This bound controls total matrix energy but gives no small angle with the vectors in (112). The distinction mirrors theorem 7.2 and corollary 7.5.

8.2 Boundary statement

The results can now be placed in a precise order:

$$
\begin{gathered}
\text{finite local model and exact row drift} \\
\Downarrow \\
\text{all source rows below } X^{1/2-o(1)} \text{ are closed} \\
\Downarrow \\
\text{one factorable } (D, L, K) \text{ wedge is closed} \\
\Downarrow \\
\text{residual energy is large, but generic phases cancel} \\
\Downarrow \\
\text{the zero-phase fixed-shift directional coefficient remains.}
\end{gathered}
\tag{116}
$$

The final arrow is not supplied by any theorem in the paper. Existing beyond-square-root distribution results require additional factorization or coefficient structure; they do not by themselves overcome sieve parity or select $h = 2$ [2, 3, 6].

9 Finite certificate and conclusion

The analytic theorems do not depend on computation. The accompanying script `experiments/square_root_gate_certificate.py` checks the finite algebra underlying the paper:

(i) the arbitrary-modulus complete-period mean in lemma 2.1;

(ii) the tail formula for $\Delta_{h,R}(m)$, including rows $m > R$;

(iii) Vaughan's identity for asymmetric parameters and arbitrary finite test arrays;

(iv) the decomposition of $\beta_V(k)$ into the visible region $\ell d \leq R$ and invisible region $\ell d > R$;

(v) the simplified drift (77);

(vi) the exponent ledger in corollary 6.4;

(vii) the direct and Fourier forms of the translation correlation in proposition 7.3.

The committed JSON records the input sizes, exact rational checks, maximum floating-point Fourier discrepancy, and a SHA-256 digest of its canonical payload. It tests conventions and finite identities only. It does not test Bombieri–Vinogradov, Maynard's theorem, an asymptotic limit, or fixed-shift cancellation.

9.1 What changed from TPC-15

TPC-15 stopped at a broad calibrated Type-II packet. The present paper adds four strict refinements.

First, a row beyond the finite-model cutoff is not treated as an unknown black box. Its complete-period model mean is explicit, and the missing Euler factors form the drift (10). This makes it possible to distinguish a genuine prime-progression error from a finite spectral truncation effect.

Second, the drift is summably small throughout the classical averaged progression range. The asymmetric choice $V = 1$ then moves the source-row boundary from $X^{1/2-\delta}$ to $X^{1/2-o(1)}$. The resulting normal form (40) is simpler than the symmetric TPC-15 packet and proves that the old left wing was not intrinsic.

Third, the symmetric coordinates remain useful after the divisor label d is retained. The exact boundary is $\ell d = R$, and Maynard's fixed-residue theorem controls one nonempty exterior wedge approaching modulus size $X^{11/21}$. This is a genuine $o(X)$ estimate for an actual piece of the hard packet. It does not control the sum over all divisor scales.

Fourth, the energy calculation rules out a misleading strategy. The Ramanujan residual has leading energy $X \log(X/R)$, so it cannot be made small merely by increasing a finite local cutoff within the present range. Translation and factor-phase identities show that the remaining task is angular: one must control a distinguished fixed-shift arithmetic direction, not merely total energy or a generic phase.

9.2 Next gate

The next layer has several honest outcomes. One may try to enlarge the Maynard-admissible three-scale polytope, seek cancellation after summing a controlled family of D-slices, or prove a sharper bound for the actual directional coherence (98). A negative result would also be informative if it exhibited a family of residual matrices with small phase averages and large zero-phase alignment under exactly the available local constraints.

What cannot be done is to infer the zero-phase estimate from the phase mean square, to identify a fixed quotient slice as ordinary Type I, or to promote one factorable subpacket to the complete balanced block. For an admissible even h, the final total condition remains

$$\mathcal{H}^{\sharp}_{h,R;M}(X) = o(X),$$

which is equivalent to the weighted Hardy–Littlewood asymptotic by theorem 4.2. No assertion is made that this condition has been proved for any prescribed h, and in particular not for $h = 2$.

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